Closed-loop object search | same initial scene, independent policies

Target: packaged_food_1   Camera: bottom   Scene: packaged_food_1_scene00027_env0008

Red: remove blocker. Green: target selected. A row is one policy decision before removal.

Object peak score -> first candidate with visibility >= 50%; step 0 is the common starting image.

Target reference

Fusion | removals: 0
Similarity | removals: 0
Occlusion | removals: 7
Complexity | removals: 15
r=0
Observed scene
Raw map (0-1)
FOUND packaged_food_1 | visible 100.0%
score 0.9953 | skipped 0
Observed scene
Raw map (0-1)
FOUND packaged_food_1 | visible 100.0%
score 0.9841 | skipped 0
Observed scene
Raw map (0-1)
REMOVE book_4 | visible 86.9%
score 0.6815 | skipped 0
Observed scene
Raw map (0-6)
REMOVE packaged_food_4 | visible 51.9%
score 3.952 | skipped 0
r=1
Finished at r=0
No further observation or action
Finished at r=0
No further observation or action
REMOVE packaged_food_3 | visible 72.0%
score 0.4055 | skipped 0
REMOVE fruit_4 | visible 100.0%
score 3.164 | skipped 1
r=2
Finished at r=0
No further observation or action
Finished at r=0
No further observation or action
REMOVE fruit_4 | visible 100.0%
score 0.3237 | skipped 0
REMOVE packaged_food_3 | visible 70.4%
score 3.313 | skipped 1
r=3
Finished at r=0
No further observation or action
Finished at r=0
No further observation or action
REMOVE toy_2 | visible 98.5%
score 0.1376 | skipped 0
REMOVE book_3 | visible 97.2%
score 2.854 | skipped 0
r=4
Finished at r=0
No further observation or action
Finished at r=0
No further observation or action
REMOVE fruit_1 | visible 100.0%
score 0.1001 | skipped 0
REMOVE fruit_1 | visible 100.0%
score 2.573 | skipped 0
r=5
Finished at r=0
No further observation or action
Finished at r=0
No further observation or action
REMOVE book_3 | visible 97.4%
score 0.09651 | skipped 0
REMOVE book_4 | visible 100.0%
score 1.967 | skipped 0
r=6
Finished at r=0
No further observation or action
Finished at r=0
No further observation or action
REMOVE packaged_food_2 | visible 100.0%
score 0.02886 | skipped 0
REMOVE fruit_3 | visible 100.0%
score 1.878 | skipped 0
r=7
Finished at r=0
No further observation or action
Finished at r=0
No further observation or action
FOUND packaged_food_1 | visible 100.0%
score 0.005818 | skipped 0
REMOVE fruit_2 | visible 99.9%
score 1.686 | skipped 0
r=8
Finished at r=0
No further observation or action
Finished at r=0
No further observation or action
Finished at r=7
No further observation or action
REMOVE toy_3 | visible 100.0%
score 1.45 | skipped 0
r=9
Finished at r=0
No further observation or action
Finished at r=0
No further observation or action
Finished at r=7
No further observation or action
REMOVE toy_2 | visible 100.0%
score 1.249 | skipped 0
r=10
Finished at r=0
No further observation or action
Finished at r=0
No further observation or action
Finished at r=7
No further observation or action
REMOVE book_1 | visible 85.2%
score 1.075 | skipped 0
r=11
Finished at r=0
No further observation or action
Finished at r=0
No further observation or action
Finished at r=7
No further observation or action
REMOVE toy_1 | visible 100.0%
score 1.047 | skipped 0
r=12
Finished at r=0
No further observation or action
Finished at r=0
No further observation or action
Finished at r=7
No further observation or action
REMOVE book_2 | visible 96.1%
score 0.8588 | skipped 0
r=13
Finished at r=0
No further observation or action
Finished at r=0
No further observation or action
Finished at r=7
No further observation or action
REMOVE toy_4 | visible 100.0%
score 0.05065 | skipped 0
r=14
Finished at r=0
No further observation or action
Finished at r=0
No further observation or action
Finished at r=7
No further observation or action
REMOVE packaged_food_2 | visible 100.0%
score 0.07519 | skipped 0
r=15
Finished at r=0
No further observation or action
Finished at r=0
No further observation or action
Finished at r=7
No further observation or action
FOUND packaged_food_1 | visible 100.0%
score 0.1002 | skipped 0

After step 0, removal choices and observed scenes can differ across policies. Blank cells mean the policy has finished.

Grayscale: fixed F/S/O 0-1, C 0-6 display cap; no per-image normalization. Contours/marks use simulator IDs for action annotation.

Simulator visibility eligibility and idealized object removal; target retrieval adds one separate action. Post-hoc extreme examples.